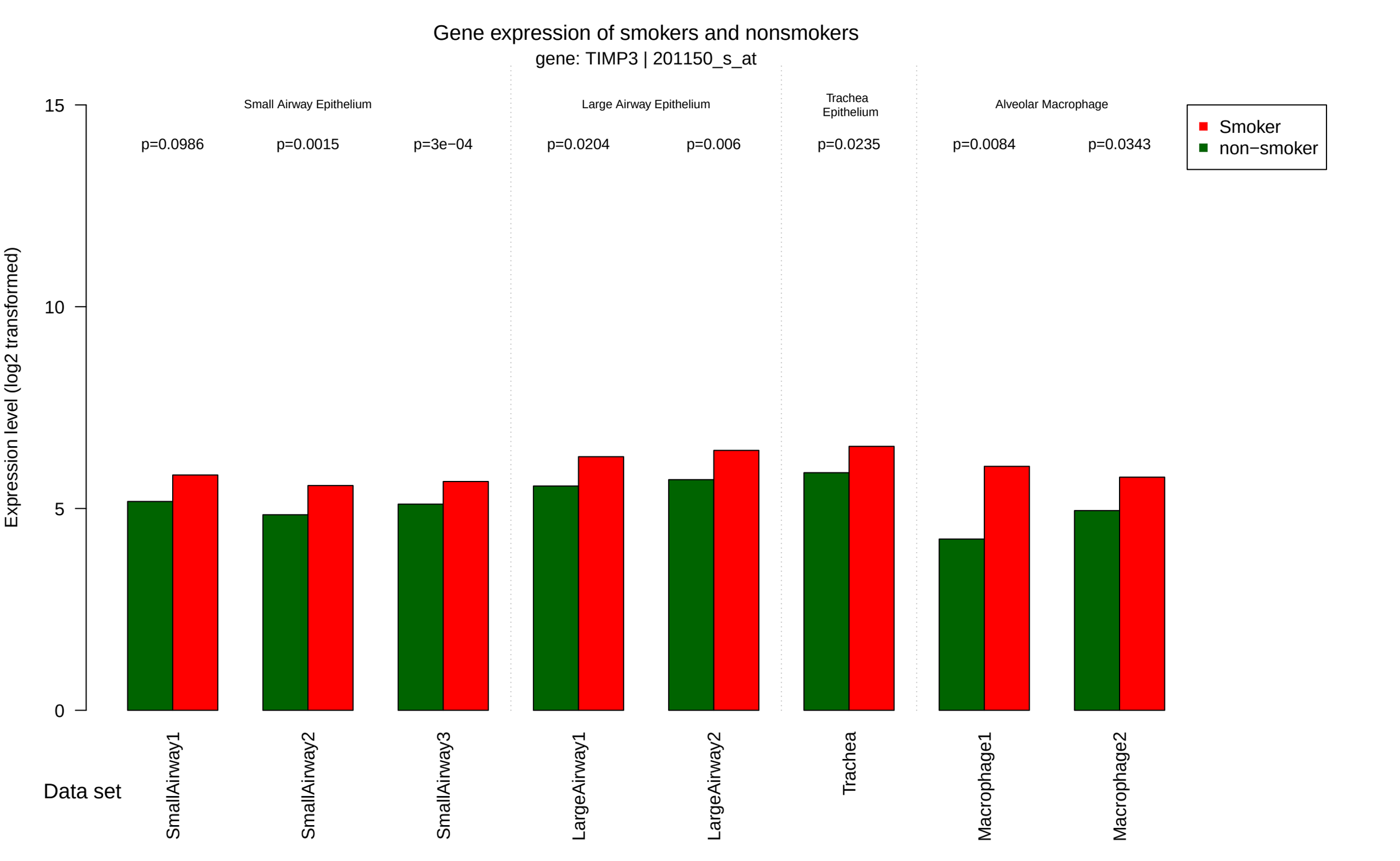
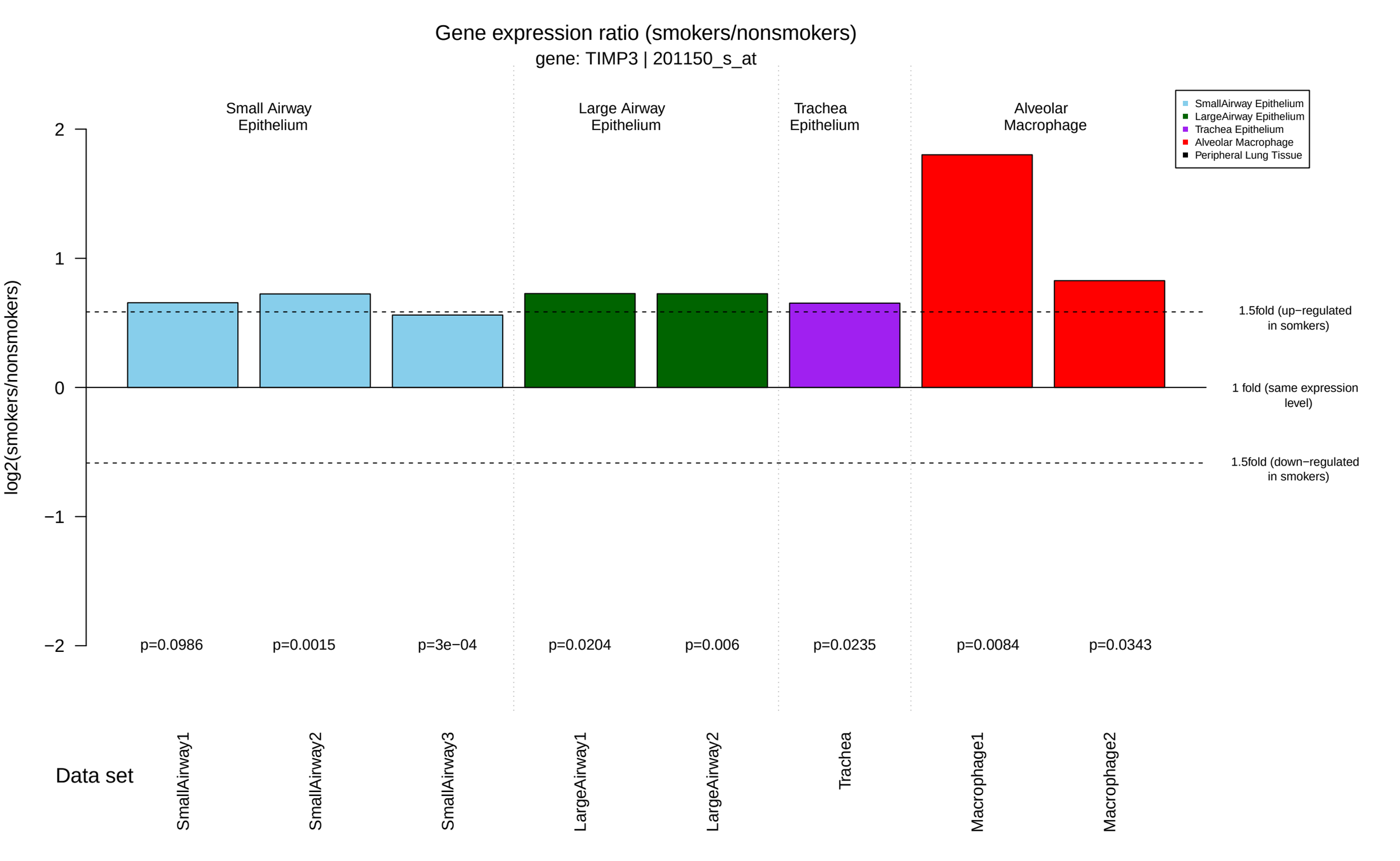
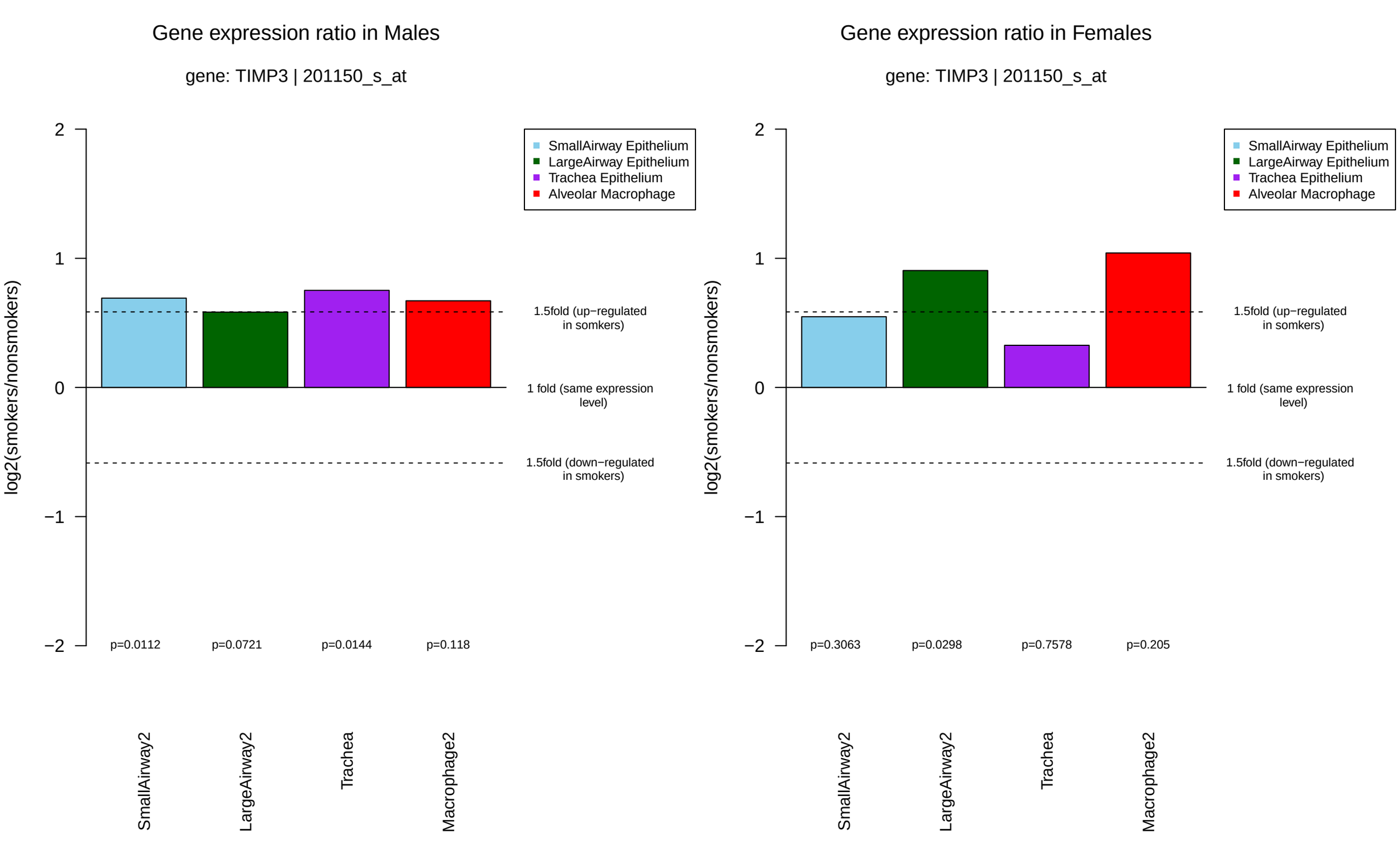
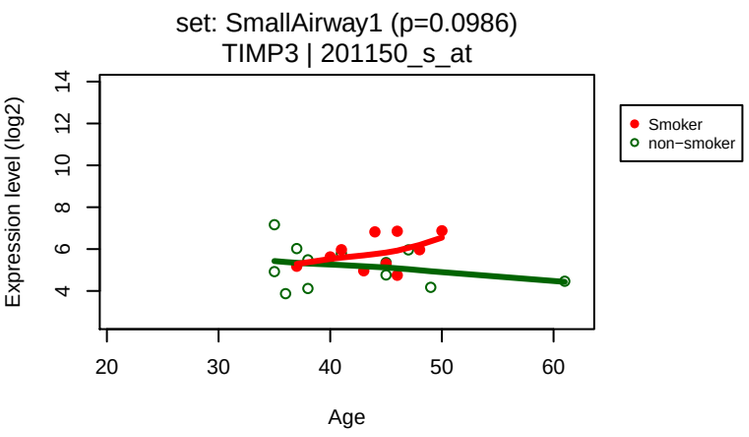
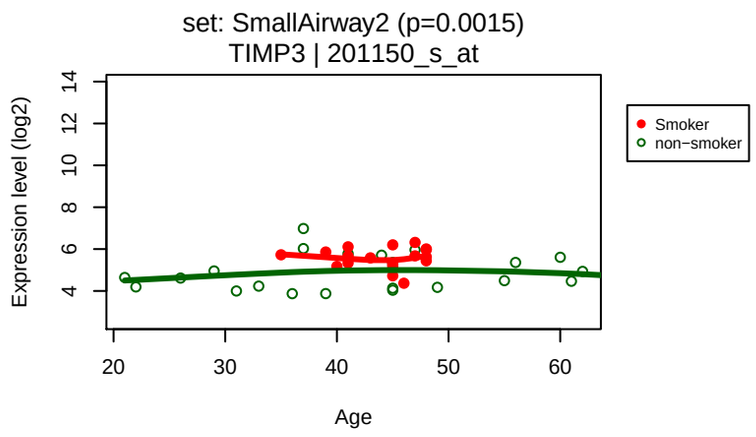
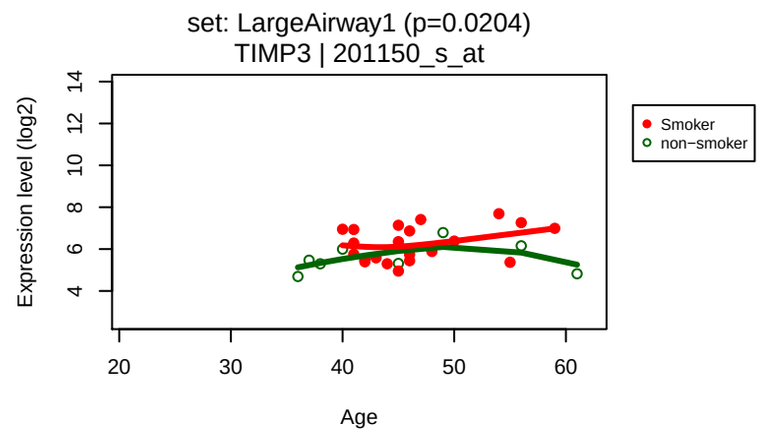

Gene expression of smokers and nonsmokers
gene: TIMP3 | 201150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0986
p=0.0015
p=3e−04
p=0.0204
p=0.006
p=0.0235
p=0.0084
p=0.0343
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

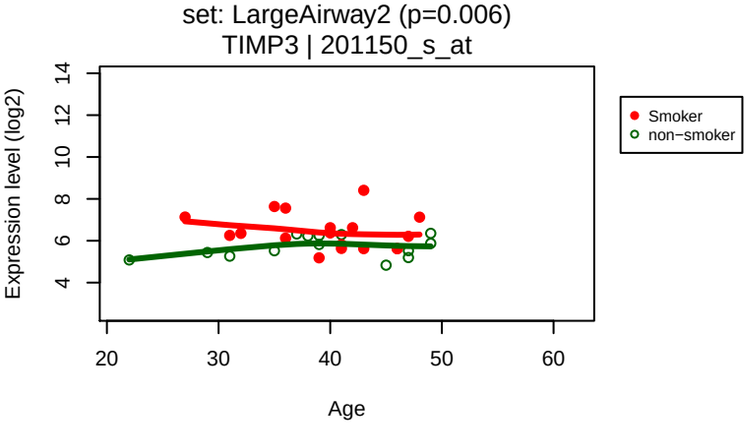

Gene expression ratio (smokers/nonsmokers)
gene: TIMP3 | 201150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0986
p=0.0015
p=3e−04
p=0.0204
p=0.006
p=0.0235
p=0.0084
p=0.0343
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

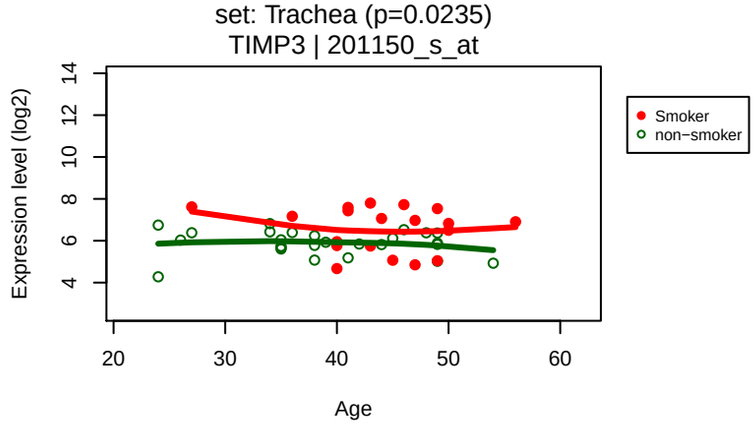

Gene expression ratio in Males
gene: TIMP3 | 201150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.0721
p=0.0144
p=0.118
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIMP3 | 201150_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3063
p=0.0298
p=0.7578
p=0.205
SmallAirway2
LargeAirway2
Trachea
Macrophage2

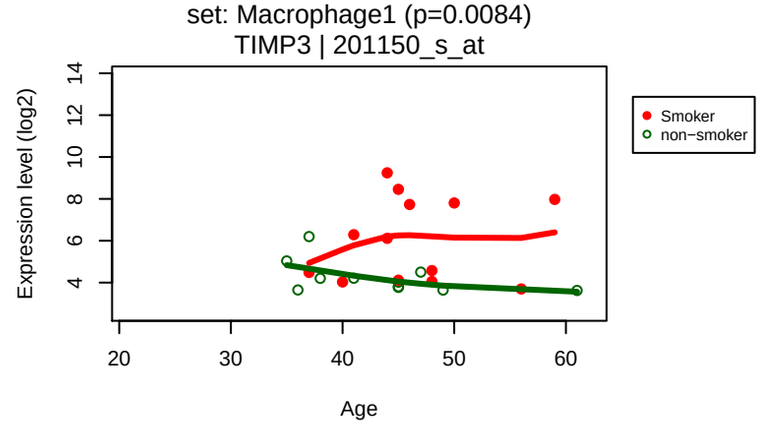

set: SmallAirway1 (p=0.0986)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker

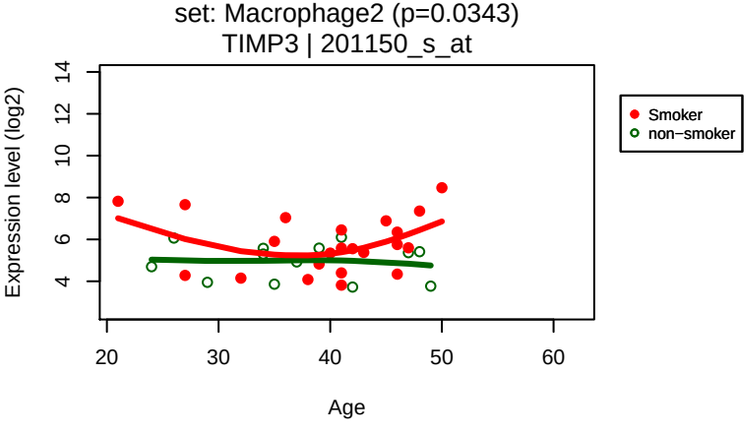

set: SmallAirway2 (p=0.0015)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker

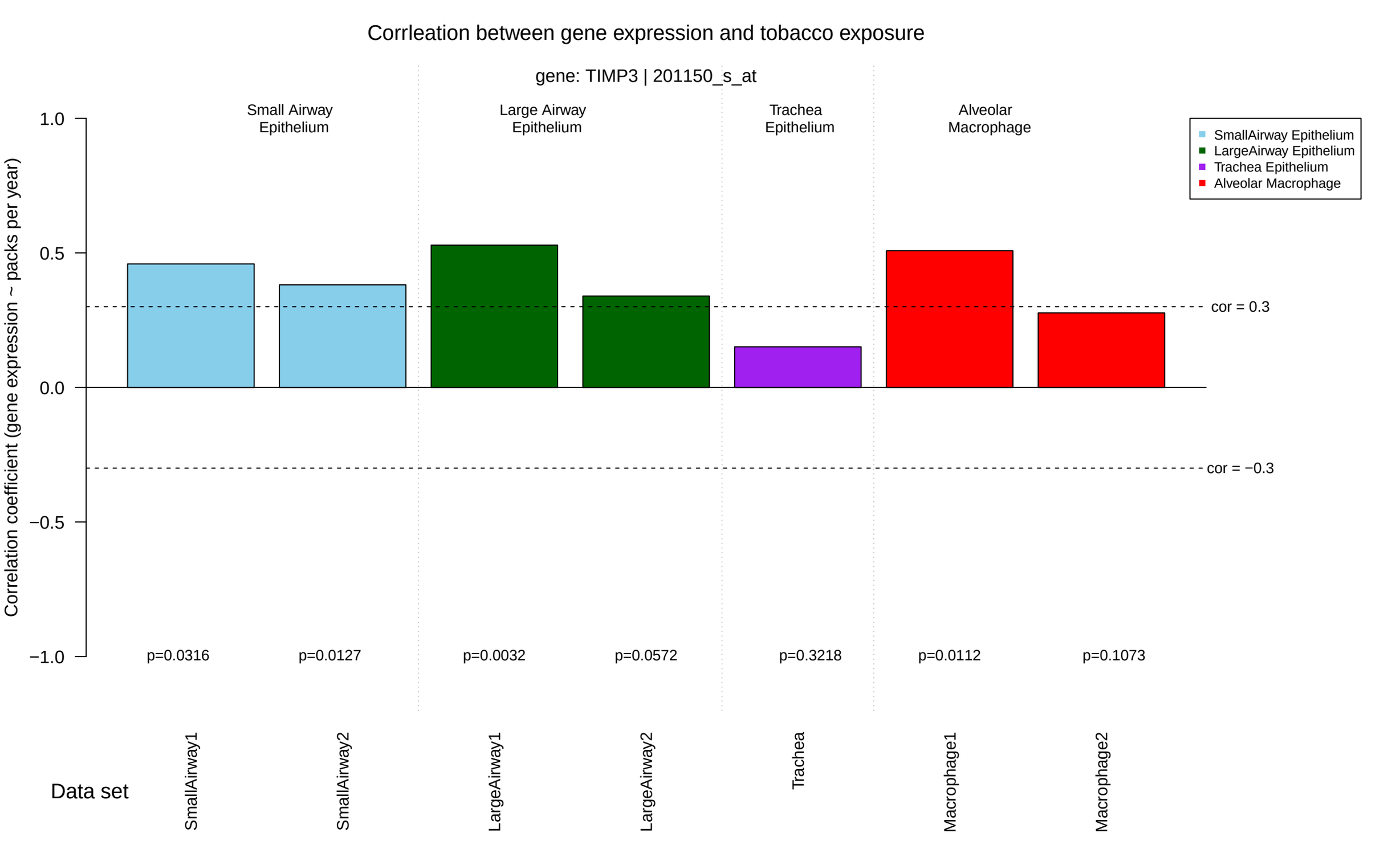

set: LargeAirway1 (p=0.0204)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.006)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0235)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0084)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0343)
TIMP3 | 201150_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIMP3 | 201150_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0316
p=0.0127
p=0.0032
p=0.0572
p=0.3218
p=0.0112
p=0.1073
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2